



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL APPLICATIONS [SET A]

FULL MARKS: 100
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets []
This question paper consists of five printed pages.*

SECTION A (40 MARKS)
(Attempt all questions from this section)

Question 1

Choose the correct answers to the questions from the given options.

[20]

(Do not copy the questions, write the correct answers only)

- (i) Locals are involved in conservation of forests along with forest officials in
(a) Agro forestry (b) Social forestry
(c) Joint Forest Management (d) Silviculture

- (ii) Bubble Theory is a concept to reduce
(a) Water Pollution (b) Air Pollution
(c) Soil pollution (d) Noise pollution

- (iii) The picture shown depicts



- (a) Zings (b) Ahar (c) Johads (d) Eris

- (iv) Identify the correct statement
(a) Subsidy is the fine paid by the polluter
(b) Subsidy is the no taxation on the non eco-friendly products
(c) Subsidy is the lower tax duties on eco-friendly technologies
(d) Subsidy is the higher tax duties on eco-friendly technologies
- (v) Select the correct statements for Fluidized Bed Combustion
1. It results in a turbulent mixing of gases and solids

2. It results in reduction of sulphur oxides only
 3. It is a costly process
 4. Limestone is used to precipitate out sulphate during combustion
- (a) 1,3 and 4 (b) 2, 3 and 4 (c) 1 and 4 (d) 2 and 4

(vi) **Assertion (A):** Curitiba in Brazil is a city known for its sustainability model.
Reason (R): The city planners of Curitiba integrated traffic management with land use planning.

- (a) A and R are true and R explains A
 (b) A and R are true but R is not the explanation of A
 (c) A is false but R is true
 (d) A is true but R is false

(vii) **Assertion (A):**Community forestry involves local communities in forest management.
Reason(R): Local community lacks knowledge about the indigenous trees and methods of conservation of forests.

- (a) A and R are true and R explains A
 (b) A and R are true but R is not the explanation of A
 (c) A is false but R is true
 (d) A is true but R is false

(viii) The vehicle which uses two or more distinct power sources

- (a) Electric vehicles (b) Hybrid vehicles
 (c) Alternative vehicle (d) Smart vehicles

(ix) 'Many urban centres have landfill sites, and often, waste is not disposed in these sites.' In this context which of the following improves solid waste management?

- (a) Teaching people the importance of segregation
 (b) Hospital waste to be gathered collectively
 (c) Composting is a bad option
 (d) Staff handling the waste should be discouraged

(x) The device which works by using a catalyst to stimulate a chemical reaction by converting to less harmful substances

- (a) Three stroke engine (b) Hybrid vehicles
 (c) Two stroke engines (d) Catalytic converters

(xi) Among the statements given below select the correct ones that describe the concept of sustainable development.

1. Planned growth with minimum damage to the environment
2. Growth irrespective of the extent of damage to the environment
3. Growth that caters to ecology, economy and society
4. Minimum developmental works in the field of ecology

- (a) 1 and 4 (b) 1 and 3 (c) 2 and 4 (d) 4 only

(xii) Identify the bus shown



- (a) Feeder bus (b) Express bus (c) Bi-articulated bus (d) Rapid bus

- (xiii) Given below are few methods of conservation of resources. Choose the correct method of soil conservation.
- (a) Metal recycling (b) Trickle drip irrigation
(c) Mulching and dry farming (d) Shifting cultivation
- (xiv) Rooftop rain water harvesting is practiced in cities for storing rainwater. Select the correct sequence of the structures for rooftop rainwater harvesting.
1. Conduit
2. Recharge structure
3. Catchment area
4. Storage tank
- (a) 1,3,2 and 4 (b) 3,1,2 and 4 (c) 2,4,3 and 1 (d) 4,2,1 and 3
- (xv) A semi solid residual of waste water from various sources is called
- (a) Manure (b) Fertilizer (c) Biomass (d) Sewage sludge
- (xvi) Building ridges and channels alternatively where crops are grown on ridges is called
- (a) Check dams (b) Wind breaks
(c) Contour bunding (d) Terrace farming
- (xvii) A process of deposition of soil particles in the waterways and dams is
- (a) Leaching (b) Siltation (c) Filtration (d) Erosion
- (xviii) _____ is a surface area that drains all the streams and rainfall to a common point.
- (a) Johad (b) Watershed (c) Wetland (d) Tankas
- (xix) Select the correct pair
- (a) Methane—Global warming
(b) Sulphur dioxide— ultraviolet radiation
(c) carbon dioxide---carboxyhaemoglobin
(d) Ozone—Acid rain
- (xx) Biomass refers to
- (a) Plant waste, animal waste and agricultural residue
(b) Industrial waste
(c) Biogas Plant
(d) Fertilizers and pesticides

Question 2

- (i) The calorific values of petrol and CNG are 45,000kJ/kg and 50,000kJ/ kg, respectively. If you have vehicle which can run on petrol as well as CNG, which fuel will you prefer and why? [2]
- (ii) State *any two* reasons to prove landfills are not environment friendly. [2]
- (iii) Why are organic fertilizers preferred over chemical fertilizers? [3]
- (iv) Name the Convention of international importance on wetlands and state its *two* objectives. [3]

Question 3

- (i) How is night soil beneficial? [2]
- (ii) How ineffective is incineration as a method of waste disposal? [2]
- (iii) State *any three* limitations of implementing Euro norms in India. [3]
- (iv) What role does recycled wood play in promoting environmental sustainability? [3]

SECTION B (60 MARKS)
(Attempt any four questions from this section)

Question 4

- (i) Study the picture and answer the questions that follow

[5]



- (a) Which city is depicted in the above picture?
(b) State *any four* practices adopted by the people which makes this city very sustainable.
- (ii) Discuss *any five* reasons for the treatment of waste water before discharge into water bodies. [5]
- (iii) As a student what personal goals would you set to conserve the environment? [5]

Question 5

- (i) Though Community participation brings lasting benefits to people, it has its own set of limitations. Comment. [5]
- (ii) Study the picture given below and answer the questions. [5]



How does the above toilet function? Justify with reasons why such toilets are gaining prominence in today's world.

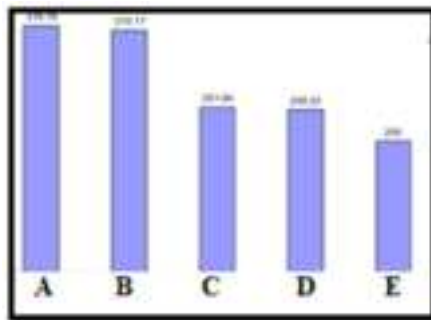
- (iii) Mr. M.C. Mehta, an environmentalist, and public interest attorney, visited The Taj Mahal in 1984 and noticed that the white marble of the Taj Mahal was turning yellow. To bring this matter into the limelight, he filed a PIL before the Supreme Court drawing the attention of the Court towards the degradation of the Taj Mahal. What initiatives were taken by the Apex Court to safeguard the national heritage? [5]

Question 6

- (i) What is Social forestry? Explain how it encompasses all the three aspects of economy, ecology and society. [5]
- (ii) Though Damodar Valley Corporation is involved in providing hydroelectric power to the surrounding areas, it comes with a heavy cost on the environment. Justify. [5]
- (iii) State *any four* contributions made by Tarun Bharat Sangh towards the conservation of the environment. Mention the name of the founder of TBS. [5]

Question 7

- (i) The graph given below shows the concentration of SPM in the atmosphere in the various locations A,B,C, D and E of a country. Study the graph and answer the questions that follow. [5]



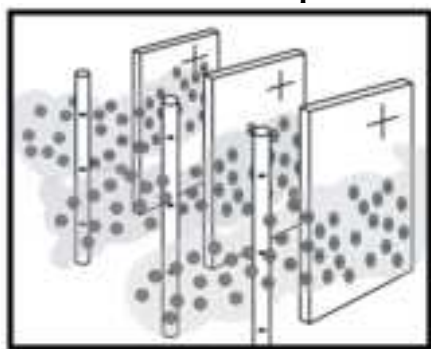
- (a) List *any two* harmful effects on health caused by SPM.
(b) Briefly explain *any four* initiatives that can be taken to reduce the emission of SPM in locations A and B.
- (ii) Describe *any one* traditional rainwater harvesting mechanism used in Rajasthan and highlight its effectiveness in desert climate. [5]
- (iii) Cite examples to explain how traditional knowledge and practices contribute to modern soil conservation efforts in rural areas. [5]

Question 8

- (i) Remote sensing satellites have played an important role in collecting information over the past few decades. Suggest *five reasons* for the popularity of this form of technology. [5]
- (ii) Suggest *five methods* by which we can combat deforestation. [5]
- (iii) The shift to alternative fuels not only supports environmental goals but also offers economic and energy security benefits. Justify. [5]

Question 9

- (i) Study the diagram given below and answer the question that follows: [5]



- (a) Identify the technological device shown in the picture.
(b) How does the above device help to reduce pollution?
- (ii) It could be said that the smell of wood smoke from the hearth is as old as humanity itself. With such a long association, the risks from wood smoke may be difficult to recognize. Although wood has a very high calorific value, we still discourage its use as a fuel. Explain. [5]
- (iii) Trisha has seen huge garbage dumps outside the school campus which are not being regularly disposed of. Next day she delivered this information in the assembly and convinced the school mates to join her for an awareness rally. [5]
- Who do you think is responsible for disposing of the garbage at regular intervals from public places? What would happen, if all the wastes produced at public places are not disposed of regularly?